Notes: Easley and O'Hara (JF, 2004)
Information and the Cost of Capital

Contents

1 Big picture

1.1 One-sentence summary

This note summarizes Easley and O'Hara (2004), which develops a multi-asset rational-expectations asset-pricing model in which the split between public and private information changes investors' perceived risk, portfolio holdings, equilibrium prices, and therefore a firm's cost of equity capital.

1.2 Central question

The paper asks a simple but important question: if market participants and regulators believe that disclosure quality, analyst coverage, and market design affect the cost of capital, why do standard asset-pricing models usually omit information structure from expected returns?

The paper's answer is that private information creates an information-risk channel. Informed investors can adjust their portfolio weights after observing private signals, while uninformed investors cannot fully infer those signals from prices. Thus uninformed investors are systematically disadvantaged: they hold too much of stocks with bad private news and too little of stocks with good private news. Because this risk cannot be diversified away in equilibrium, investors demand a higher return for stocks with more private information.

1.3 Core economic mechanism

The mechanism can be written as a causal chain:

More private information $\Rightarrow$ larger informed-uninformed information gap $\Rightarrow$ different portfolio weights $\Rightarrow$ extra risk borne by uninformed traders $\Rightarrow$ higher required return.

Equivalently, firms with more public, precise, and widely disseminated information face a lower cost of capital.

1.4 Why this paper matters

The paper connects three areas that are often treated separately:

- **Asset pricing:** expected returns are affected by information risk, not only covariance with aggregate risk.
- **Corporate finance:** firms can influence their own cost of capital through disclosure, accounting quality, analyst coverage, and listing venue.
- **Market microstructure:** price informativeness matters not only for trading efficiency but also for equilibrium expected returns.

2 Incremental contribution of the paper

2.1 Contribution relative to standard CAPM-style pricing

In a standard CAPM world, investors share the same information and hold the same tangency portfolio. Idiosyncratic asset-level risks should be diversified away. Easley and O'Hara show that this conclusion changes when investors have asymmetric information. The market portfolio remains mean-variance efficient only with respect to average beliefs, but no actual trader necessarily wants to hold it. Therefore, information risk can be priced even though it is asset-specific.

2.2 Contribution relative to rational-expectations information models

Earlier rational-expectations models such as Grossman and Stiglitz (1980), Admati (1985), and Wang (1993) analyze informed trading and price informativeness. Easley and O'Hara shift the focus to the firm's cost of capital. Their key innovation is to compare public and private information directly and show that, holding total information constant, replacing public information with private information raises the required return.

2.3 Contribution relative to investor-recognition models

Merton (1987) shows that limited investor recognition can raise the cost of capital because fewer investors know about an asset. Easley and O'Hara assume that all investors know every stock exists, but some investors know more about its payoff. Thus the problem is not awareness of the asset; it is asymmetric information about value. This distinction matters because the information premium in Easley and O'Hara cannot be mechanically arbitraged away by tilting toward high-return stocks.

2.4 Contribution for corporate policy

The paper provides a formal rationale for why corporate disclosure, accounting precision, analyst following, and trading venue can affect valuation. These decisions do not necessarily alter product-market cash flows, but they alter the information environment in which investors price the firm's equity.

3 Model structure

3.1 Assets and payoffs

The economy has two dates: today, when investors trade, and tomorrow, when payoffs are realized. There is one risk-free asset with price normalized to one and K risky stocks indexed by $k = 1, \ldots, K$. The future value of stock k is

$$v_k \sim N(\bar{v}_k, \rho_k^{-1}),$$

where ρ_k is the precision of the prior distribution. The per-capita supply of stock k is random:

$$x_k \sim N(\bar{x}_k, \eta_k^{-1}).$$

Random supply is a standard device in rational-expectations models. Economically, it can be interpreted as noise trading or random float supply.

3.2 Signals: public and private information

For each stock k, there are I_k signals about v_k. Each signal is conditionally independent and normally distributed around the true payoff:

$$s_{ki} \mid v_k \sim N(v_k, \gamma_k^{-1}).$$

A fraction α_k of the signals is private, and a fraction $1 - \alpha_k$ is public. The paper defines two sufficient statistics:

$$N_k = \frac{1}{\alpha_k I_k} \sum_{i=1}^{\alpha_k I_k} s_{ki}, \tag{1}$$

$$M_k = \frac{1}{(1-\alpha_k)I_k} \sum_{i=\alpha_k I_k+1}^{I_k} s_{ki}. \tag{2}$$

Here N_k summarizes private signals and M_k summarizes public signals. Their precisions are

$$\alpha_k I_k \gamma_k \quad \text{and} \quad (1-\alpha_k) I_k \gamma_k,$$

respectively. Varying α_k changes the split between public and private information while keeping the total number of signals fixed.

3.3 Informed and uninformed investors

A fraction μ_k of investors receives the private signals about stock k. All investors observe public signals. The key difference is therefore:

- **Informed investors** observe both M_k and N_k.
- **Uninformed investors** observe M_k and the equilibrium price p_k, but not N_k directly.

All investors have CARA utility with risk-aversion parameter $\delta > 0$.

4 Investor demand and Bayesian updating

4.1 Generic portfolio demand

Suppose investor j believes that stock k has conditional mean $\bar{v}_k^j$ and precision ρ_k^j. With CARA-normal preferences, the investor solves a mean-variance problem. His demand for stock k is

$$z_k^j = \frac{\bar{v}_k^j - p_k}{\delta(\rho_k^j)^{-1}}. \tag{3}$$

The demand is increasing in perceived expected excess payoff $\bar{v}_k^j - p_k$ and decreasing in perceived variance $(\rho_k^j)^{-1}$ and risk aversion δ.

4.2 Demand of informed investors

An informed investor observes all I_k signals. Bayes' rule gives

$$\bar{v}_k^I = \frac{\rho_k \bar{v}_k + \gamma_k \sum_{i=1}^{I_k} s_{ki}}{\rho_k + \gamma_k I_k}, \tag{4}$$

$$\rho_k^I = \rho_k + \gamma_k I_k. \tag{5}$$

Therefore the informed investor's demand is

$$z_k^I = \frac{\rho_k \bar{v}_k + \gamma_k \sum_{i=1}^{I_k} s_{ki} - p_k(\rho_k + \gamma_k I_k)}{\delta}. \tag{6}$$

Economic interpretation: informed investors update strongly because they see the private signals directly. When private news is good, they buy more; when private news is bad, they reduce holdings or short.

4.3 Demand of uninformed investors

Uninformed investors do not observe the private signal statistic N_k. They infer it imperfectly from the equilibrium price. The paper conjectures a linear price function of the form

$$p_k = av_k + b\sum_{i=1}^{\alpha_k I_k} s_{ki} + c\sum_{i=\alpha_k I_k+1}^{I_k} s_{ki} - dx_k + e\bar{x}_k. \tag{7}$$

The price is informative because informed demand embeds private signals, but it is not fully revealing because random supply/noise trading also affects price. The uninformed investor therefore constructs a signal θ_k from price and public information. This signal has mean v_k and precision $\rho_{\theta k}$.

The uninformed investor's posterior is based on prior information, public signals, and the price-derived signal:

$$\bar{v}_k^U = \frac{\rho_k \bar{v}_k + \gamma_k \sum_{i=\alpha_k I_k+1}^{I_k} s_{ki} + \rho_{\theta k}\theta_k}{\rho_k + \gamma_k(1-\alpha_k)I_k + \rho_{\theta k}}, \tag{8}$$

$$\rho_k^U = \rho_k + \gamma_k(1-\alpha_k)I_k + \rho_{\theta k}. \tag{9}$$

Since $\rho_k^U < \rho_k^I$ whenever private information is not fully revealed by price, uninformed investors face higher perceived risk.

5 Rational-expectations equilibrium

5.1 Market clearing

For each stock, per-capita supply equals weighted demand from informed and uninformed investors:

$$\mu_k z_k^I + (1-\mu_k)z_k^U = x_k. \tag{10}$$

Solving this equation yields a partially revealing rational-expectations equilibrium.

5.2 Equilibrium price

The paper shows that there exists an equilibrium price of the form

$$p_k = av_k + b\sum_{i=1}^{\alpha_k I_k} s_{ki} + c\sum_{i=\alpha_k I_k+1}^{I_k} s_{ki} - dx_k + e\bar{x}_k. \tag{11}$$

The important economic content is not the algebraic coefficients themselves, but the fact that price aggregates private information imperfectly. Private information reaches uninformed investors only through the noisy price signal.

6 The cost of capital result

6.1 Expected return per share

The expected return per share for stock k is

$$\mathbb{E}[v_k - p_k] = \frac{\delta\bar{x}_k}{\rho_k + (1-\alpha_k)I_k\gamma_k + \mu_k\alpha_k I_k\gamma_k + (1-\mu_k)\rho_{\theta k}}. \tag{12}$$

This is the central pricing expression of the paper.

6.2 Reading the denominator

Equation (12) says that the required return falls when the stock's payoff is perceived more precisely. The denominator contains four parts:

1. ρ_k: prior precision about the stock.
2. $(1 - \alpha_k)I_k\gamma_k$: precision contributed by public signals.
3. $\mu_k\alpha_k I_k\gamma_k$: precision of private signals weighted by the fraction of informed investors.
4. $(1 - \mu_k)\rho_{\theta k}$: price-inferred precision available to uninformed investors.

Public information directly improves everyone's posterior precision. Private information improves precision for informed traders, but it disadvantages uninformed traders unless it is sufficiently revealed through price.

6.3 Why more private information raises the required return

The paper's main comparative static is

$$\frac{\partial \mathbb{E}[v_k - p_k]}{\partial \alpha_k} > 0 \quad \text{if } \mu_k < 1. \tag{13}$$

Holding total information constant, moving signals from public to private increases the required return. The logic is as follows:

- Public signals are observed by everyone, so they reduce uncertainty symmetrically.
- Private signals are observed only by informed investors, so they create a portfolio advantage for informed investors.
- Uninformed investors cannot fully reconstruct private signals from prices because prices also contain supply noise.
- Therefore uninformed investors view the asset as riskier and demand compensation.

6.4 Why this is not simple idiosyncratic risk

The information premium is not eliminated by holding many stocks. Uninformed investors are always on the wrong side of private information: they receive less of good-news stocks and more of bad-news stocks relative to informed traders. The problem is not just volatility; it is adverse portfolio selection.

7 Portfolio intuition and mean-variance efficiency

7.1 Informed and uninformed traders hold different portfolios

Informed investors condition directly on private information; uninformed investors only infer it imperfectly. Thus they choose different holdings. On average, the informed hold more of risky stocks because they face less uncertainty. The paper derives

$$\mathbb{E}[Z_k^I - Z_k^U] = \delta^{-1}\mathbb{E}[v_k - p_k](\rho_k^I - \rho_k^U) > 0. \tag{14}$$

The size of this gap increases with the amount of private information.

7.2 Why the CAPM intuition fails

In the CAPM, everyone has the same information and the same risk-aversion coefficient, so everyone holds the same market portfolio. Here, investors share the same utility form but not the same information set. Hence the usual separation theorem fails: each investor's optimal portfolio is mean-variance efficient conditional on his information, but those efficient portfolios are not the same.

7.3 Market portfolio and average beliefs

The paper shows that the market portfolio is mean-variance efficient with respect to average conditional beliefs. However, no actual representative agent with those beliefs necessarily exists. Thus the market portfolio can be efficient in a formal average-belief sense while still not being optimal for any real investor.

7.4 Connection to Rock's IPO underpricing logic

The information-risk channel resembles Rock (1986). In IPOs, informed investors buy more when the issue is good and avoid bad issues; uninformed investors are left with a worse allocation. Here, the logic extends to secondary markets and all stocks: informed investors tilt their portfolio toward good private-news stocks and away from bad private-news stocks, while uninformed investors cannot perfectly imitate them.

8 Comparative statics and corporate implications

8.1 Fraction of informed traders

Let μ_k be the fraction of investors who receive private information. The paper shows

$$\frac{\partial \mathbb{E}[v_k - p_k]}{\partial \mu_k} < 0. \tag{15}$$

More informed investors reduce the cost of capital. There are two channels:

- More informed investors directly increase demand because the asset is less risky for them.
- Their aggregate trading makes price more informative for uninformed investors.

This provides a rationale for analyst coverage and institutional following. Even if analysts do not make information public perfectly, increasing the dispersion of information can lower the risk premium.

8.2 Precision of information

Let γ_k be the precision of signals. The paper shows

$$\frac{\partial \mathbb{E}[v_k - p_k]}{\partial \gamma_k} < 0. \tag{16}$$

More precise information lowers the cost of capital. This is where accounting quality matters: accounting choices do not necessarily change real cash flows, but they change the precision of investors' information, which affects valuation.

8.3 Prior precision and firm maturity

Let ρ_k be the precision of the prior. The paper shows

$$\frac{\partial \mathbb{E}[v_k - p_k]}{\partial \rho_k} < 0. \tag{17}$$

Established firms may have lower costs of capital because investors already know more about them. Young firms and IPO firms face higher costs of capital because prior precision is low and much of the available information is private.

8.4 When there is no public information

The paper also highlights a subtle point: when no public information exists, some private information can be better than no information at all. If $\alpha_k = 1$, then

$$\frac{\partial \mathbb{E}[v_k - p_k]}{\partial I_k} < 0. \tag{18}$$

This is because informed trading makes the price more informative. However, conditional on total information being fixed, public information is better than private information from the perspective of lowering the cost of capital.

8.5 Disclosure, analysts, and market design

The model implies that firms can affect their own cost of capital by changing their information environment. Examples include:

- more transparent disclosure that turns private signals into public signals;
- higher-quality accounting that increases signal precision;
- analyst coverage that either discovers or disseminates information;
- listing and trading venue choices that affect price informativeness.

The paper therefore gives an asset-pricing foundation for the idea that disclosure policy and microstructure are corporate-finance variables.

9 Tables and figures from the paper

9.1 Table I: Numerical Example - The Effect of Parameters on Excess Return

Table I
Numerical Example: The Effect of Parameters on Excess Return

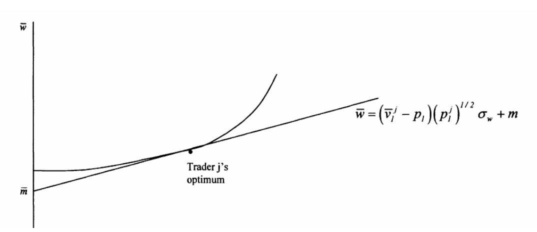
This table illustrates the percentage change in expected excess return ($\%\Delta ER_k$) for stock k generated by changes in the fraction of signals that are private (α_k), the fraction of traders who are informed (μ_k), and the precision of signals (γ_k). In each panel, the parameters other than the parameter of interest are fixed at $\rho_k = \gamma_k = \eta_k = 1$, $\delta = 1$, $\bar{x}_k = 1$, $I_k = 10$, $\mu_k = 0.2$ and $\alpha_k = 0.5$.

Panel A											
α_k	0		0.2		0.4		0.6		0.8		1.0
$\%\Delta ER_k$		15.6		15.5		15.6		15.9		16.4	
Panel B											
μ_k	0		0.2		0.4		0.6		0.8		1.0
$\%\Delta ER_k$		−27.8		−21.7		−10.2		−4.6		−2.2	
Panel C											
γ_k			0.8		0.9		1.0		1.1		1.2
$\%\Delta ER_k$				−11.3		−10.3		−9.5		−8.8	

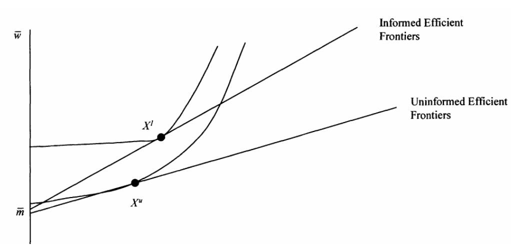
Table I: Numerical example on how private information, informed-trader dispersion, and signal precision affect expected excess return.

Interpretation. Table I gives numerical magnitudes for the model. Increasing the private-information share α_k increases the risk premium; increasing the fraction of informed traders μ_k lowers it; increasing signal precision γ_k also lowers it. The largest effects in this calibration come from the dispersion and precision of information, showing why analyst following and accounting quality can matter economically.

9.2 Figure 1: The Efficient Frontier for Trader j and Figure 2: Efficient Frontiers for Informed and Uninformed Traders



(a) Figure 1: Trader-specific efficient frontier.



(b) Figure 2: Informed versus uninformed frontiers.

Extracted figures on trader-specific mean-variance frontiers.

Interpretation. Figure 1 shows that each investor chooses the tangent point between his perceived efficient frontier and indifference curve. Figure 2 shows that informed investors face a better frontier because they have greater posterior precision. As a result, informed and uninformed investors optimally choose different risky holdings even when they share the same risk-aversion coefficient.

9.3 Figure 3: The Effect of Good and Bad News on Portfolios of Informed and Uninformed Traders

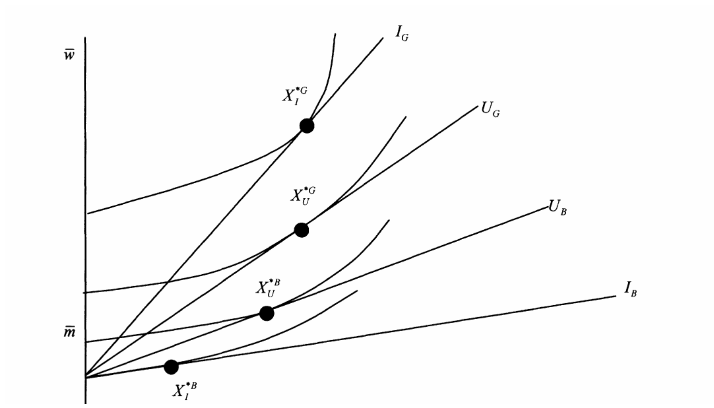


Figure 3: Private news changes the portfolio gap between informed and uninformed investors.

Interpretation. Figure 3 illustrates the asymmetry created by private signals. When news is good, informed investors increase their risky holdings more than uninformed investors. When news is bad, informed investors reduce risky holdings more sharply. Therefore uninformed investors are systematically disadvantaged by not being able to condition directly on the private signal.

10 How to read the paper's equations

10.1 The key expression to remember

The most important equation is the required-return expression:

$$\mathbb{E}[v_k - p_k] = \frac{\delta \bar{x}_k}{\rho_k + (1-\alpha_k)I_k\gamma_k + \mu_k\alpha_k I_k\gamma_k + (1-\mu_k)\rho_{\theta k}}.$$

Everything in the paper can be read as changing either the numerator or denominator.

10.2 Numerator: why risk and supply matter

The numerator is $\delta \bar{x}_k$.

- Higher risk aversion δ means investors require more compensation.
- Higher per-capita supply $\bar{x}_k$ means more risky shares must be absorbed by investors.

If there is no average supply of the risky asset, no one must bear its risk on average, and the premium disappears.

10.3 Denominator: why information lowers risk

The denominator is total effective precision. Increasing prior precision, public signal precision, private-information dispersion, or price informativeness all lowers the required return.

However, shifting information from public to private reduces the effective precision available to uninformed investors and increases the adverse portfolio-selection problem. This is why the sign of $\partial \mathbb{E}[v_k - p_k]/\partial \alpha_k$ is positive.

11 Empirical and research implications

11.1 Direct empirical implication

The most direct empirical implication is:

More information-based trading risk $\Rightarrow$ higher expected return / higher cost of capital.

This is the logic developed empirically in Easley, Hvidkjaer, and O'Hara (2002), which uses microstructure estimates of the probability of informed trading.

11.2 Connection to disclosure research

The paper predicts that better disclosure should reduce the cost of capital, especially for firms with low analyst following or high information asymmetry. The key empirical difficulty is separating disclosure quality from omitted firm characteristics, but the theory is clear: disclosure lowers the cost of capital by turning private information into public information or by raising signal precision.

11.3 Connection to analyst coverage

Analyst coverage can lower the cost of capital in two ways. First, analysts may create new information, raising γ_k. Second, they may disseminate information, raising the effective fraction of informed traders μ_k or making information closer to public. Either channel lowers the required return in the model.

11.4 Connection to IPOs and young firms

New firms have low prior precision ρ_k and often high private-information intensity. The model therefore predicts a high risk premium for IPOs and young firms. This provides an information-risk explanation for why small or newly listed firms may have high expected returns.

11.5 Connection to insider-trading regulation

The model implies two competing effects. Some private information can make prices more informative relative to no information. But holding total information fixed, converting private information into public information reduces the required return. Therefore insider-trading enforcement can lower the cost of capital if it shifts information from private channels into public channels without sharply reducing total information production.

12 Limitations and interpretation

12.1 Model limitations

The model is deliberately stylized. It uses a two-period CARA-normal rational-expectations setup, exogenous information acquisition, and random supply. These assumptions allow clean equilibrium expressions but abstract from dynamic information production, strategic disclosure, endogenous analyst behavior, and multiple rounds of trading.

12.2 Why the simplification is useful

The simplification isolates the portfolio channel. Private information matters not because it mechanically increases volatility, but because it lets informed investors choose better portfolio weights. This channel is difficult to see in standard symmetric-information pricing models.

12.3 What should not be overinterpreted

The paper does not say that all private information is bad. It says that, holding the total amount of information fixed, public information is better for reducing the cost of capital. When the comparison is private information versus no information, private information may lower the cost of capital by making prices more informative.

13 Short summary

Easley and O'Hara (2004) build a multi-asset rational-expectations model in which information structure affects expected returns. When more information is private rather than public, informed investors can adjust their portfolios more effectively than uninformed investors. Uninformed investors are left with an adverse allocation of good-news and bad-news stocks, so they require a higher expected return. The result links asset pricing to corporate disclosure, accounting quality, analyst coverage, market microstructure, and IPO information environments.

14 Conclusion

The paper's central message is that information is a priced determinant of the cost of capital. More private information raises the risk premium because it creates an adverse portfolio-selection problem for uninformed investors. More public information, more precise information, and wider information dissemination lower the cost of capital. This provides a theoretical foundation for why firms care about accounting quality, disclosure policy, analyst coverage, and trading venues even when these choices do not change real cash flows directly.